

DIABETES RISK ANALYSIS PROJECT - MATH 108C

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Abstract. In this project, we will construct a model to predict diabetes risk percentage. We will train this model by utilizing diabetes datasets from the CDC, considering factors such as education, age, income, and various health indicators. The goal is to provide the model with a set of factors about an individual, and receive an estimation of diabetes risk for this individual. From there, we will apply this model across geographic areas, based on census data, and generate predicted values for each area. Using these predicted values by location, we will construct a heatmap of predicted diabetes risk, and compare these results to real world diabetes dispersion data.

1. Introduction

In this project, we aim to predict the risk of diabetes using demographic data by applying a sequence of data analysis and dimensionality reduction techniques. Our approach is motivated by principal component analysis, which we will utilize for dimensionality reduction. We begin by centering the dataset and analyzing the structure of the centered dataset using the Gram matrix, which captures relationships between variables. To extract the most informative patterns from this matrix, we use the QR algorithm, a reliable numerical method for computing eigenvalues and eigenvectors. These eigenvalues allow us to then obtain a Singular value decomposition, which will form the foundation of Principal Component Analysis (PCA). This will allow us to reduce the dataset's dimensionality while preserving the most significant variation in the data. To ensure that each feature contributed comparably to the analysis, we normalized the data after centering by scaling each feature to have unit variance. This pipeline—from Gram matrix to QR algorithm to SVD to PCA allows us to explore how well basic demographic features can explain and predict geographic trends in diabetes prevalence.

We will then present a variety of maps of predicted diabetes risk by state, and compare this to real world results and data. We will conclude with an analysis of our results, accuracy, errors, and future directions.

1.1. Dataset Overview: CDC Diabetes Health Indicators. Our project draws on the *CDC Diabetes Health Indicators Dataset*, published through the UCI Machine Learning Repository and originally derived from the 2015 Behavioral Risk Factor Surveillance System (BRFSS). The dataset includes 253,680 entries, each representing an individual surveyed by the CDC. Its purpose is to explore associations between lifestyle, health history, and diabetes prevalence in the United States.

The target variable is `Diabetes_binary`, where:

- 0 = No diabetes
- 1 = Prediabetic or Diagnosed diabetes

The dataset includes 19 other predictive features across demographic, behavioral, and medical domains, described in the following table.

Table 1.1: Summary of Selected Variables from the CDC Diabetes Dataset

Variable Name	Type	Description / Coding
HighBP	Binary	1 = High blood pressure diagnosis, 0 = No
HighChol	Binary	1 = High cholesterol, 0 = No
CholCheck	Binary	1 = Cholesterol checked in last 5 years, 0 = No
BMI	Integer	Body Mass Index (self-reported)
Smoker	Binary	1 = Smoked 100+ cigarettes, 0 = No
Stroke	Binary	1 = Diagnosed stroke, 0 = No
HeartDiseaseAttack	Binary	1 = Coronary heart disease or heart attack, 0 = No
PhysActivity	Binary	1 = Any physical activity in past 30 days, 0 = No
Fruits	Binary	1 = Eats fruit ≥ 1 /day, 0 = No
Veggies	Binary	1 = Eats vegetables ≥ 1 /day, 0 = No
HvyAlcoholConsump	Binary	1 = Heavy drinker (men >14 /wk, women >7 /wk), 0 = No
GenHlth	Integer (1–5)	1 = Excellent, ..., 5 = Poor
MentHlth	Integer (0–30)	Days mental health was not good (past 30 days)
PhysHlth	Integer (0–30)	Days physical health was not good (past 30 days)
DiffWalk	Binary	1 = Difficulty walking or climbing stairs, 0 = No
Sex	Binary	0 = Female, 1 = Male
Age	Integer (1–13)	1 = 18–24, ..., 13 = 80+
Education	Integer (1–6)	1 = No schooling, ..., 6 = College graduate
Income	Integer (1–8)	1 = $<10k$, ..., 8 = $\$75k+$

Before release, the dataset underwent preprocessing including bucketing of age into categorical groups and the removal of entries with missing values. As a result, all variables are either binary indicators or integer-coded categorical responses, and the final dataset contains no missing data.

The wide range of health and lifestyle factors makes this dataset particularly rich for modeling and analysis. In our project, we focus primarily on the demographic and behavioral features that are readily available at the state level for simulation and prediction.

2. Eigenvalues of the Gram Matrix via the Real Symmetric QR Algorithm

To deepen our understanding of dimensionality reduction and prepare for principal component analysis (PCA), we investigate the Gram matrix associated with our data. Given a mean-centered data matrix $A \in \mathbb{R}^{n \times p}$ with n observations and p features, the *Gram matrix* is defined as

$$G := A^T A.$$

This matrix encodes the inner products between the columns of A , revealing how variables are linearly related. Since G is real, symmetric, and positive semi-definite, it has real, non-negative eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p \geq 0$, and an orthonormal basis of eigenvectors $\{v_1, v_2, \dots, v_p\}$. These eigenvalues and eigenvectors form the theoretical foundation for PCA: the eigenvalues indicate the amount of variance explained along each direction, and the eigenvectors define those directions.

Let Q be the matrix whose columns are the eigenvectors of G , and let Λ be the diagonal matrix of eigenvalues. Then we have the eigenvalue decomposition:

$$G = Q \Lambda Q^T.$$

Projecting the data matrix A onto the eigenvectors of G yields the principal components:

$$Z = A Q.$$

Each row z_i of Z is a p -dimensional vector representing the i -th observation in the new coordinate system formed by the eigenvectors. The total variance captured by the first k principal components is proportional to $\sum_{j=1}^k \lambda_j$.

In our project, this process helps us identify the most informative variables in the diabetes dataset. Features corresponding to large eigenvalues are those that vary the most and tend to be more linearly independent. Thus, PCA using the Gram matrix provides a principled way to reduce dimensionality while preserving the structure and variability of the data. This will be discussed further in the PCA section.

Computing Eigenvalues via the Real Symmetric QR Algorithm. The QR algorithm is one of the most effective numerical techniques for computing all eigenvalues of a matrix. For general matrices, the algorithm must address complex eigenvalues and perform careful shift strategies. However, for real symmetric matrices like the Gram matrix G , the QR algorithm enjoys several advantages:

- All eigenvalues of a real symmetric matrix are guaranteed to be real.
- Symmetric matrices can be reduced to tridiagonal form, which is preserved under QR iterations, making the process both faster and more stable.

The QR algorithm generates a sequence of orthogonally similar matrices that converge to a diagonal matrix of eigenvalues. Given $A_0 = G$, each iteration proceeds as:

$$\begin{aligned} A_k - \mu_k I &= Q_k R_k, \\ A_{k+1} &= R_k Q_k + \mu_k I, \end{aligned}$$

where Q_k is orthogonal, R_k is upper triangular, and μ_k is a shift (often chosen using the Wilkinson shift for faster convergence). All matrices A_k are similar to G , so they share its eigenvalues.

Before applying the QR algorithm, we first reduce G to symmetric tridiagonal form using a unitary similarity transformation. This preprocessing step makes each QR iteration cheaper to compute and allows the algorithm to converge more rapidly.

Once convergence is reached, we recover the eigenvalue decomposition

$$G = Q \Lambda Q^T,$$

where Λ contains the eigenvalues and Q the corresponding orthonormal eigenvectors. This decomposition allows us to extract meaningful features from our data, perform PCA, and guide variable selection in a principled and interpretable way.

Reliability of the Gram Matrix and Eigenvalue computation. During our analysis, we relied on the Gram matrix $G = A^T A$ as part of our Singular value decomposition (SVD) and Principal Component Analysis (PCA) pipeline. However, after our presentation, it was noted that computing eigenvalues via the Gram matrix can be numerically unstable, particularly when the original data matrix A is ill-conditioned or when the dataset has highly correlated features. This is because forming $A^T A$ can significantly amplify rounding errors and lead to overestimated eigenvalues, especially for smaller singular values, thereby distorting the variance captured by the principal components.

To address this concern and verify the reliability of our results, we tested multiple computational approaches. In particular, we compared the eigenvalues computed from the Gram matrix using the QR algorithm with results from numpy's more stable built-in symmetric eigenvalue solver `np.linalg.eigvalsh()`, applied directly to the centered covariance matrix. Both approaches produced the same singular values and principal components, confirming that our results were consistent and that, in our specific case, the numerical stability of the Gram matrix was not a limiting factor.

Nevertheless, we acknowledge that relying solely on the Gram matrix can be problematic in other contexts, especially with noisy or high-dimensional data.

2.1. Computational Results. The eigenvalues of the Gram matrix computed via the QR algorithm were:

$$\begin{bmatrix} 2.87 \times 10^{-13} & 2.19 \times 10^2 & 2.43 \times 10^2 & 3.66 \times 10^2 & 4.22 \times 10^2 & 4.46 \times 10^2 & 4.67 \times 10^2 \\ 7.85 \times 10^2 & 9.01 \times 10^2 & 1.05 \times 10^3 & 1.10 \times 10^3 & 1.41 \times 10^3 & 1.59 \times 10^3 & 1.88 \times 10^3 \\ 4.65 \times 10^3 & 6.52 \times 10^3 & 3.73 \times 10^4 & 6.86 \times 10^4 & 1.66 \times 10^5 & 3.08 \times 10^5 & 4.73 \times 10^6 \end{bmatrix}$$

3. Singular Value Decomposition

3.1. Mathematical Motivation. The spectral theorem provides a powerful decomposition for symmetric matrices. SVD provides a similar decomposition that is applicable to any matrix (even if they are not square!).

Essentially, SVD revolves around the fact that the image of a matrix A on the unit ball (which determines the whole space by scaling) is an ellipsoid of dimension $\text{rank}(A)$. Then, the matrix can be viewed as rotating the unit ball to the (orthogonal) axes of the ellipsoid, scaling them, and then rotating the ellipsoid to the image; this is the Singular value decomposition.

The set of singular values of a linear matrix $A : V \rightarrow W$ are the square roots of the Gram matrix A^*A listed in decreasing order and as many times as the multiplicity of the corresponding eigenvalue of A^*A . Since A^*A is a symmetric square matrix, the set of singular values is exactly as large as the rank of the matrix, and in fact there exists an orthonormal basis of V and W , $[e_i]$ and $[f_i] = A[e_i]$ so that if V and W are expressed in terms of these bases, the transformation is just the singular values dotted with the basis. In terms of matrices, this means that when $\text{rank}(A) = r$, $A = BDC^*$ where D is $r \times r$ diagonal, B is $n \times r$ orthonormal, and C is $r \times m$ orthonormal.

Finding the singular values and left singular vectors is equivalent to finding the maximum direction of norm change, since one will notice that the axes of the ellipsoid are the local maximums of the distance from the origin.

This is done by finding the vectors that maximize the value of the transformation on the unit ball and are orthogonal to all previous vectors. This clearly produces an orthonormal basis of V ; then the singular values are the scaling factors on each of these vectors, so that $Av_i = s_i u_i$ which becomes $AV = SU$ and we have the SVD if $[Av_i]$ are pairwise orthogonal. But this must be true because for $j > i$, $[Av_i, Av_j]$, if they were not orthogonal, a negligible change of v_i along the direction of v_j results in a non-negligible change of Av_i along the direction of Av_j (since it increases orthogonality and therefore increases maximization, but this contradicts the maximality of Av_i).

By definition, we calculate the singular values by calculating the square roots of the eigenvalues of the Gram matrix. However, in practice this often leads to significant error in smaller eigenvalues, as outlined in section 2. This can be avoided by constructing the symmetric matrix

$$\begin{bmatrix} 0 & A^T \\ A & 0 \end{bmatrix}$$

whose positive eigenvalues are exactly the singular values of A . Finding the eigenvalues of this matrix instead of the Gram matrix gives more accurate singular values.

3.2. Computational Results. From the eigenvalues of the Gram matrix, we know the singular values immediately to be

$$\begin{bmatrix} 5.35 \times 10^{-7} & 14.8 & 15.58 & 19.13 & 20.55 & 21.13 & 21.6 \\ 28.02 & 30.02 & 32.47 & 33.1 & 37.49 & 39.92 & 43.41 \\ 68.19 & 80.76 & 193.15 & 261.94 & 407.69 & 554.68 & 2173.8 \end{bmatrix}$$

For the sake of brevity, we shall only list out the first few eigenvectors of the Gram matrix (those associated with the largest singular values): recall that these are the elements of V in the $A = U\Sigma V^*$ decomposition. Since the elements of the U matrix are of dimension 253,680, we shall refrain from showing them here.

In order to make sense of these singular vectors, this is the order in which the features are listed:

Feature Names (in order):

$$\begin{bmatrix} \text{HighBP} & \text{HighChol} & \text{CholCheck} & \text{BMI} & \text{Smoker} & \text{Stroke} & \text{HeartDiseaseorAttack} \\ \text{PhysActivity} & \text{Fruits} & \text{Veggies} & \text{HvyAlcoholConsump} & \text{AnyHealthcare} & \text{NoDocbcCost} & \text{GenHlth} \\ \text{MentHlth} & \text{PhysHlth} & \text{DiffWalk} & \text{Sex} & \text{Age} & \text{Education} & \text{Income} \end{bmatrix}$$

Right singular vector (Singular value: $\sigma = 407.69$):

$$\begin{bmatrix} -2.18\text{e-}01 & -4.62\text{e-}03 & 8.03\text{e-}04 & -8.23\text{e-}03 & -2.80\text{e-}03 & 1.15\text{e-}02 & 1.99\text{e-}02 \\ -5.75\text{e-}03 & 2.04\text{e-}02 & 2.31\text{e-}02 & 4.63\text{e-}04 & 2.63\text{e-}02 & -2.59\text{e-}02 & 2.22\text{e-}03 \\ -5.59\text{e-}02 & 5.34\text{e-}02 & 7.33\text{e-}02 & 9.32\text{e-}01 & -1.65\text{e-}01 & 7.64\text{e-}02 & -1.90\text{e-}01 \end{bmatrix}$$

Right singular vector (Singular value: $\sigma = 554.68$):

$$\begin{bmatrix} -2.18\text{e-}01 & -1.69\text{e-}02 & 2.54\text{e-}03 & -4.13\text{e-}02 & 2.60\text{e-}02 & 4.57\text{e-}02 & 3.33\text{e-}04 \\ -2.30\text{e-}02 & 2.70\text{e-}02 & -8.39\text{e-}02 & 2.91\text{e-}02 & -4.70\text{e-}02 & 6.35\text{e-}02 & -1.84\text{e-}01 \\ -2.58\text{e-}02 & -8.74\text{e-}01 & -3.43\text{e-}01 & 2.38\text{e-}03 & -5.75\text{e-}03 & 9.79\text{e-}02 & -8.29\text{e-}02 \end{bmatrix}$$

Right singular vector (Singular value: $\sigma = 2173.8$):

$$\begin{bmatrix} -2.18\text{e-}01 & -5.68\text{e-}03 & -3.01\text{e-}03 & -2.75\text{e-}03 & 7.55\text{e-}03 & 3.29\text{e-}02 & 4.21\text{e-}03 \\ -1.94\text{e-}02 & 1.55\text{e-}02 & -7.04\text{e-}03 & 2.03\text{e-}02 & -2.30\text{e-}02 & 1.97\text{e-}02 & 1.49\text{e-}02 \\ 9.37\text{e-}02 & 4.18\text{e-}01 & -8.50\text{e-}01 & -4.21\text{e-}02 & -9.76\text{e-}03 & 1.59\text{e-}01 & -1.23\text{e-}01 \end{bmatrix}$$

4. Principal Component Analysis

4.1. Mathematical Motivation. For our project, due to the nature of our large data sets, we will first need to utilize dimensionality reduction before we are able to work with our data. One possible method for dimensionality reduction is Principal Component Analysis. This method aims to find a suitable lower dimensional subspace that can approximate our data.

Here, note that subspaces always pass through the origin, so we will need to shift our data so that the mean is at the origin, before we are able to determine a subspace. Once we center the points, we want subspace $\mathcal{V}$ such that each shifted data point is well approximated by a vector in $\mathcal{V}$. Say we would like to reduce to k dimensions. We want an orthonormal set of basis vectors $\{v_1, v_2, \dots, v_k\}$ to construct our subspace. Once we have our subspace, we will orthogonally project our shifted data onto the subspace using vectors in $\mathcal{V}$. Finally, we will shift the approximated data back.

We approach this process with two goals, maximizing the variance, and minimizing the reconstruction error. We will focus on maximizing variance, which will ultimately also minimize reconstruction error. We will find our basis vectors through an iterative process.

Let us briefly explain this process. Say that we have data points $\{x_1, x_2, \dots, x_m \in \mathbb{R}^n\}$, with mean vector

$$\mu = \frac{1}{m} \sum_{i=1}^m x_i.$$

We first shift our data so that the mean is at the origin. Our shifted data set is $\{x_i - \mu\}$. Let us consider the matrix of centered data:

$$\begin{bmatrix} x_1^T - \mu^T \\ x_2^T - \mu^T \\ \vdots \\ x_m^T - \mu^T \end{bmatrix} = \begin{bmatrix} x_1^T \\ x_2^T \\ \vdots \\ x_m^T \end{bmatrix} - \begin{bmatrix} \mu^T \\ \mu^T \\ \vdots \\ \mu^T \end{bmatrix} = A - e\mu^T$$

We have written our matrix of centered data in simpler way, where A is the matrix of our original data, and e is a vector of ones of length m . Our problem now becomes, finding a unit vector v such that

$$v = \underset{\|v\|_2=1}{\operatorname{argmax}} \frac{1}{m} \|(A - e\mu^T)v\|_2^2$$

We can use SVD to solve this! Letting σ_1 be the largest singular value of centered data matrix $A - e\mu^T$. Then we know σ_1^2 is the max value of $\frac{1}{m} \|(A - e\mu^T)v\|_2^2$, and, letting v_1 be the unit vector that attains σ_1^2 , we have our first basis vector, called the first principal component.

To continue in our iterations, the next vector we look for should solve the same maximization problem, with the extra condition of being orthogonal to v_1 ,

$$v = \underset{\substack{\|v_2\|_2=1 \\ v_2 \perp v_1}}{\operatorname{max}} \frac{1}{m} \|(A - e\mu^T)v_2\|_2^2$$

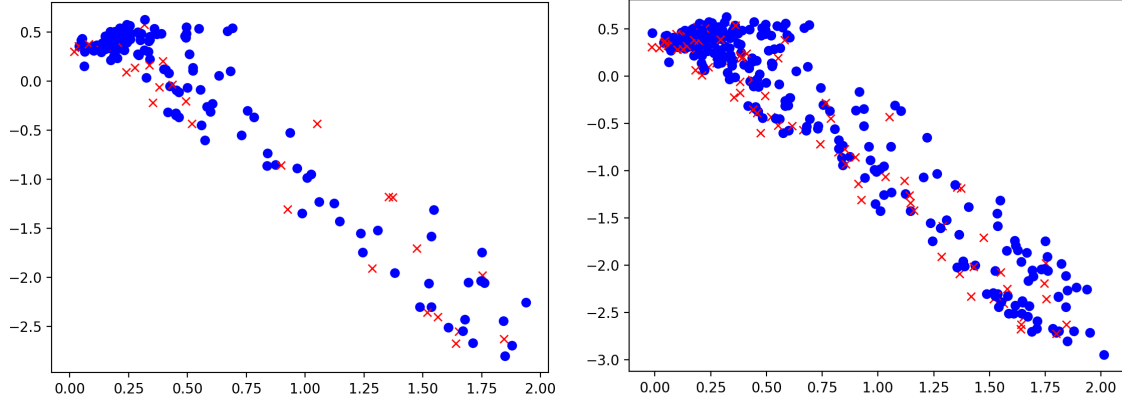


Fig. 4.1: 2-dimensional data in the coordinates of the principal components. Shown with $k=200$ and $k=500$ data points.

Again, this is solved by σ_2^2 , where σ_2 is the second largest singular value of $A - e\mu^T$. The unit vector that achieves this maximum is then v_2 , called the second principal component.

For each j , σ_j^2 solves the maximization problem, where σ_j is the j -th largest singular value of $A - e\mu^T$. The unit vector that achieves this maximum is called v_j , the j -th principal component.

Once we have our principal components $[v_1, v_2, \dots, v_k]$, we create matrix with these vectors as columns and call this

$$V_k = \begin{bmatrix} | & | & \dots & | \\ v_1 & v_2 & \dots & v_k \\ | & | & \dots & | \end{bmatrix}$$

We can now project our centered data onto this subspace, and these coordinates are given by

$$z_i = V_k^T(x_i - \mu) \in \mathbb{R}^k$$

These are called *representations*. If we write z_i as rows of matrix we call Z_k , we can note that $A - e\mu^T = Z_k V_k^T$, so our reconstruction of the original data is given by $Z_k V_k^T + e\mu^T$. We have successfully represented our data in a lower dimension k . We can choose k based on the original size of our dataset and our desired accuracy.

If we have a singular value decomposition of our centered data matrix, $A - e\mu^T = U\Sigma V^T$. The first k columns of V give our first k principal components.

4.2. Computational Results. We used PCA to reduce the dimension of our data to visualize it. Since there are too many data points to visualize, we plotted 2d data for 200 and 500 data points using projections onto the subspace spanned by the first 2 singular vectors, and 3d data for 200 data points using projections onto the subspace spanned by the first 3 singular vectors.

The results of SVD guided the selection of $k = 2$ or better $k = 3$ components to project onto balancing interpretability with retained variance. Although the largest singular value is quite large, the next 2 are also respectable in size. Adding a second dimension to the data also allows us to show more points without over-crowding.

In these figures, most of the points are blue: those are non-diabetic people. Those in red are diabetic. It is not clear that there is a separation between the diabetic and non-diabetic classes from these representations, which reflects the nature of diabetes: someone may have the exact same conditions as another, and yet develops diabetes while the other does not.

Also of note in these figures is a larger cluster closer to the origin and the rest of the points trailing off. Recall that the largest singular vector has a rather large entry of -0.85 in the difficulty walking (DiffWalk) category, and in the second singular vector DiffWalk had a relatively large -0.343 . In particular, most responses to the difficulty walking (DiffWalk) variable were a 0, with a trailing few responding with a 1. This variable

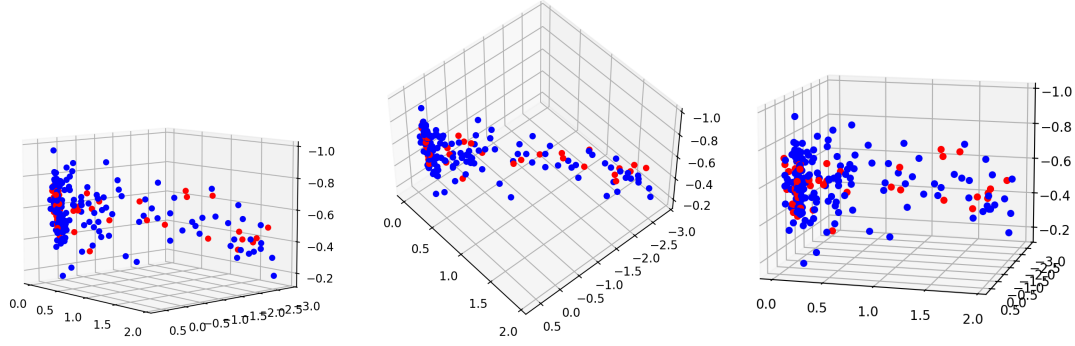


Fig. 4.2: 3 different angles of the 3 dimensional data shown in the coordinates of the principal components. k=200 data points

would have been shifted negligibly but scaled very strongly due to its low variance when normalizing the data. This leads the data to have a high density near the origin due to all the people who put a 0 and a large trail of people with a 1 in some categories leading to a much higher number when scaled for the variance. This explains the strong central behavior and large trail, which is especially visible in 2 dimensions. Once we move into 3 dimensions and the impact from DiffWalk is lessened in the third singular vector, we see less of this behavior.

4.3. Feature Scoring via Between/Within Class Variance. As part of our in-class feedback, we were asked to "Pick each feature separately. Split the data into the three classes (no diabetes, pre-diabetes, diabetes). Compute the ratio between-group variance/within-group variance. This is like an importance score for each feature. The highest value, the most important feature because it has the clearest class separation. Then, you can pick the top-k features. I think that to create a risk factor, you need to apply multinomial logistic regression to those features."

Principal Component Analysis (PCA) identifies directions of maximal variance but does not consider the class labels in the data. To quantify how useful each feature is for separating classes (in our case: No Diabetes, Prediabetes, Diabetes), we adopt a supervised scoring approach based on comparing between-group variance to within-group variance.

The idea is simple: a feature is useful for classification if the means of different classes differ significantly (high between-class variance) and the data within each class is tightly clustered (low within-class variance).

For each feature x , we compute:

$$\text{Score}(x) = \frac{\text{Between-Class Variance}}{\text{Within-Class Variance}},$$

where:

- The between-class variance is defined as:

$$\sum_{c=1}^C n_c (\mu_c - \mu)^2$$

with μ_c the mean of class c , μ the global mean, and n_c the number of data points in class c .

- The within-class variance is:

$$\sum_{c=1}^C \sum_{i \in c} (x_i - \mu_c)^2.$$

This ratio serves as an importance score for each feature: high values suggest that a feature helps clearly separate the classes. We compute this score for each feature independently, rank the features, and select the top k to be used in a predictive model.

Once the top features are identified, we apply multinomial logistic regression to them. This allows us to predict class membership (no diabetes, prediabetes, or diabetes) while maintaining interpretability through model coefficients.

4.4. Computational Results: Feature Scoring and Multinomial Logistic Regression. We applied the between-class variance to within-class variance ratio to each feature in the dataset to assess its effectiveness at distinguishing between diabetes classes. The features with the highest ratios are the most important for class separation. The results are shown below:

GenHlth	0.0943
HighBP	0.0744
DiffWalk	0.0501
BMI	0.0493
HighChol	0.0418
Age	0.0325
HeartDiseaseorAttack	0.0324
PhysHlth	0.0302
Income	0.0276
Education	0.0157

We selected the top $k = 5$ features based on this ranking: GenHlth, HighBP, DiffWalk, BMI, and HighChol. These features were used as input for a multinomial logistic regression model.

After standardizing the selected features, we fit a logistic regression model to predict diabetes class. Although our target variable was binary in this implementation (0 = no diabetes, 1 = diabetes), we used the multinomial setting for interpretability and future extension to the three-class version (no diabetes, prediabetes, diabetes).

The resulting model coefficients for the five features were:

Class 0	GenHlth	HighBP	DiffWalk	BMI	HighChol
	0.289	0.249	0.047	0.166	0.171

These coefficients quantify the contribution of each standardized feature to the predicted log-odds of diabetes status. The strongest predictors were GenHlth (general health) and HighBP (high blood pressure diagnosis), aligning with known clinical indicators of diabetes risk. Features like DiffWalk and BMI also played meaningful roles in the model.

This supervised selection and modeling approach provides a more interpretable pathway toward estimating diabetes risk based on the most relevant health indicators, and offers a foundation for constructing a simplified risk scoring system in future work.

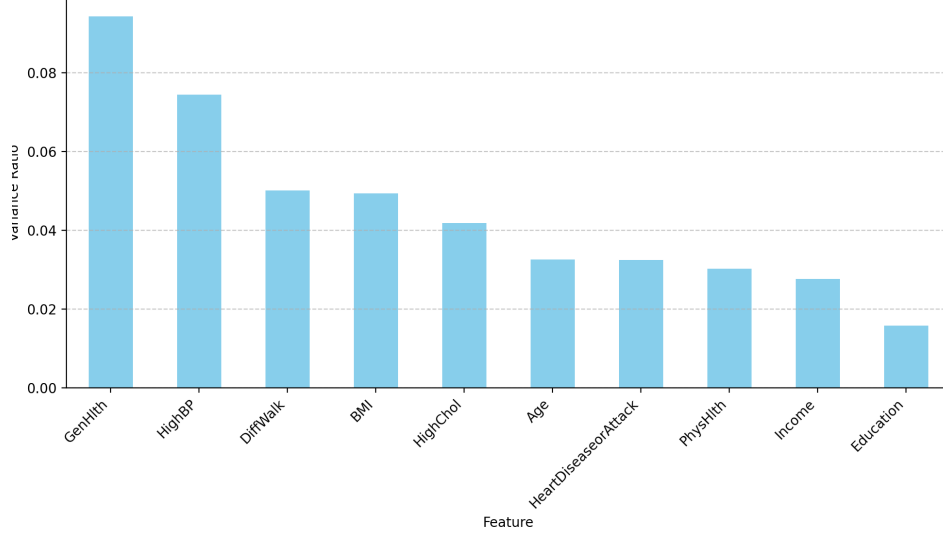


Fig. 4.3: Top 10 features ranked by the ratio of between-group to within-group variance. Higher scores indicate stronger class separation.

4.5. Future Improvement: LDA Algorithm. While PCA allowed us to reduce the dimensionality of the data by finding directions of maximal variance, it is an unsupervised method that does not consider class labels, which in our case, is diabetes status (No Diabetes, Prediabetes or Diabetes). In contrast, Linear Discriminant Analysis (LDA) is a supervised technique that aims to find directions that best separate known classes.

LDA works by projecting the data onto a lower-dimensional space in a way that maximizes class separation. Specifically, it seeks to maximize the ratio of between-class variance to within-class variance. Suppose we have a dataset of n samples $\{x_i, y_i\}$, where each $x_i \in \mathbb{R}^d$ is a feature vector and each $y_i \in \{1, 2, \dots, C\}$ is a class label. Let:

- $\mu \in \mathbb{R}^d$ be the mean of all data points,
- $\mu_c \in \mathbb{R}^d$ be the mean of class c ,
- n_c be the number of samples in class c .

Then the within-class scatter matrix is defined as:

$$S_W = \sum_{c=1}^C \sum_{x_i \in \text{class } c} (x_i - \mu_c)(x_i - \mu_c)^T,$$

and the between-class scatter matrix is defined as:

$$S_B = \sum_{c=1}^C n_c (\mu_c - \mu)(\mu_c - \mu)^T.$$

LDA seeks a projection matrix $W \in \mathbb{R}^{d \times (C-1)}$ that maximizes the "Fisher criterion":

$$W^* = \arg \max_W \det \left((W^T S_W W)^{-1} W^T S_B W \right).$$

Which is an optimization problem is equivalent to solving the generalized eigenvalue problem:

$$S_B w = \lambda S_W w.$$

The top $C - 1$ eigenvectors $w_1, w_2, \dots, w_{C-1}$ corresponding to the largest eigenvalues form the columns of the projection matrix W . Each original data point $x \in \mathbb{R}^d$ can then be projected onto this new space via:

$$z = W^T x \in \mathbb{R}^{C-1}.$$

For our case, since there are two classes (non-diabetic and prediabetic or diabetic), LDA would produce at most one discriminant axes. These axes are specifically chosen to enhance class separation, which could improve both visualization and classification performance.

Additionally, the entries of the eigenvectors w_i provide insight into the contribution of each original feature. The squared entries w_{ij}^2 quantify how strongly feature j contributes to the i -th discriminant axis. This allows us to rank features by their importance in distinguishing between classes.

Although LDA was not implemented in our current project due to time constraints, it represents a promising future direction. It is particularly well-suited for problems like ours, where the goal is to separate labeled classes.

5. Predictive Mapping with Reduced Model

To visualize geographic trends in diabetes prevalence, we used our reduced-dimension model to estimate diabetes risk by U.S. state. The goal was to see whether key demographic features alone could reasonably approximate known diabetes risk patterns.

We simulated a representative individual for each state using publicly available statistics. Due to limitations in linking individual-level medical records with geographic location, we restricted our profile to three variables:

- Education level was drawn from U.S. Census data and mapped to a 6-point scale, from 1 (no schooling) to 6 (college graduate).
- Income was estimated using per capita personal income from the BEA and mapped to an 8-point scale, from 1 (less than \$10,000) to 8 (more than \$75,000).
- BMI was approximated using state-level self-reported average weight and average height using the standard BMI formula:

$$\text{BMI} = \frac{\text{Weight (kg)}}{\text{Height (m)}^2}$$

These simulated profiles were input into a simplified linear prediction model. The model used the following regression formula to compute estimated diabetes risk:

$$\text{Diabetes Risk} = 0.3681011 + 0.02420949 \cdot \text{BMI} - 0.02074718 \cdot \text{Education} - 0.02225264 \cdot \text{Income}$$

To ensure consistency with our model assumptions, education and income levels were rescaled to match the training data:

- Education: scaled from 1 (no schooling) to 6 (college graduate)
- Income: scaled from 1 (less than \$10,000) to 8 (\$75,000+)

Each state's simulated profile was used to predict a single diabetes risk value, which was then visualized across the U.S. in the form of a heatmap.

5.1. Map Visualizations and Comparison. We constructed two heatmaps for visualization:

1. Actual diabetes risk by state - compiled from county-level diabetes rates from the Rural Health Info Hub. We computed a population-weighted average using the formula:

$$\text{State Rate} = \frac{\sum (\text{County Pop} \times \text{County Diabetes } \%) }{\sum (\text{County Pop})}$$

2. Predicted diabetes risk by state – generated by feeding simulated profiles into our reduced regression model.

The comparison revealed broad similarity in risk distribution across regions, though some discrepancies remain due to simplification.

To further evaluate model performance, we computed the absolute difference between predicted and actual rates by state and visualized the errors.

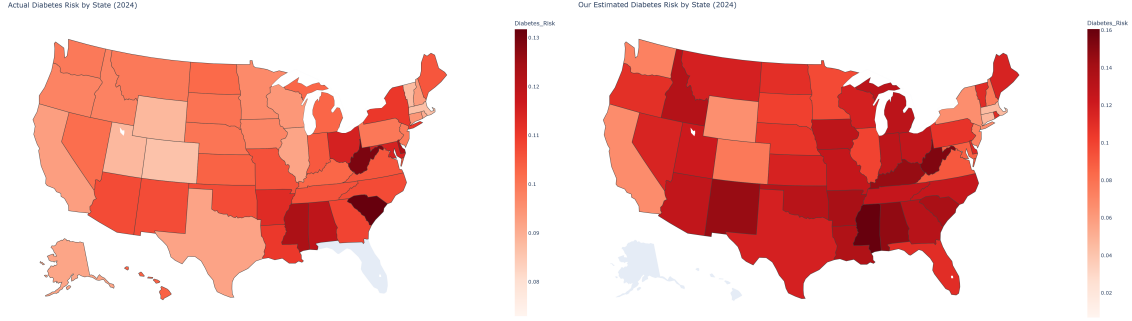


Fig. 5.1: Left: Actual Diabetes Risk by State (2024). Right: Predicted Diabetes Risk from PCA-based Model.

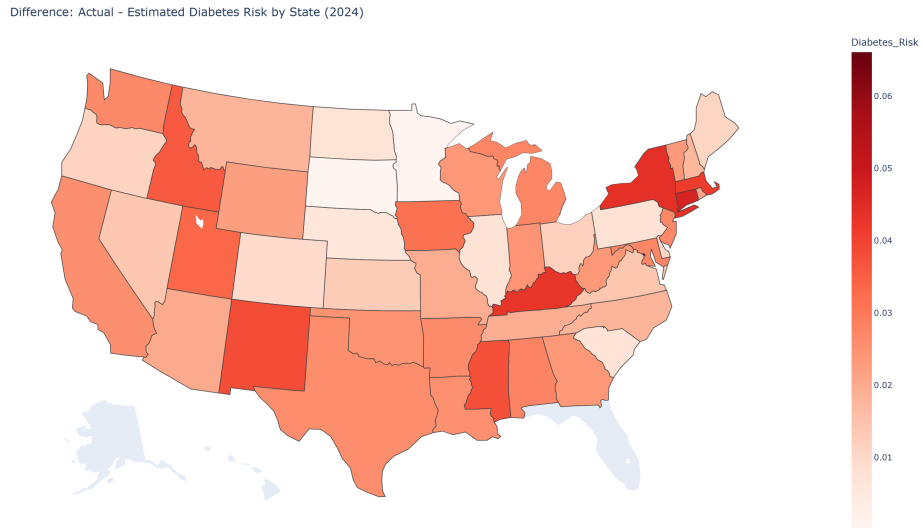


Fig. 5.2: Comparison of absolute errors between predicted and actual diabetes risk by state. Bars reflect $|\text{Predicted} - \text{Expected}|$ values for each state.

6. Process and Error

6.1. Modeling Limitations and Error Sources. While our results indicate that PCA can reveal major drivers of diabetes risk, the predictive model has inherent limitations:

- **Simplified Simulation:** We used only one representative profile per state. This ignores within-state variability.
- **Data Mapping:** Translating categorical variables (e.g., education, income) to numerical values involved simplifications that may have introduced bias.
- **Nonlinearity:** PCA captures linear structure. However, diabetes risk likely depends on complex non-linear relationships among variables.

Although the model was useful in producing a state-level prediction of diabetes risk, it is important to note its inherent limitations. As discussed in Section 4.3, the features of Income, Education, and BMI which were selected using the between-class to within-class variance ratio had relatively low scores overall. This indicates that the classes (e.g., diabetic vs. non-diabetic) are not strongly separable based on any of those individual feature, which limits the discriminative power of the model. As such, we do not expect the prediction performance to be exceptional. Nonetheless, the model served an important role in allowing us to link individual-level risk factors to broader geographical trends, which was the primary motivation behind this approach.

7. Conclusion

In this study, we applied Principal Component Analysis (PCA) to reduce the dimensionality of a health data set from UC Irvine, enabling visualization and exploration of patterns across diabetic and non-diabetic populations, and eventually generating a model to predict diabetes risk percentages.

Our PCA-reduced model was further applied to simulate diabetes risk across U.S. states using demographic proxies for education, income, and BMI. The comparison between actual diabetes prevalence data and PCA-based predictions revealed a broadly consistent spatial pattern, particularly highlighting known high-risk regions such as the Southeast. This demonstrates the potential utility of low-dimensional projections for geographic health risk mapping. However, the approach involved several simplifications, including the linear assumptions of PCA and the numeric encoding of categorical variables, which may introduce bias and overlook nonlinear relationships critical to diabetes risk. Moreover, the model’s inability to capture genetic predispositions, healthcare access, and other unmeasured variables was reflected in some inconsistent individual predictions. These limitations are mainly due to lack of data, particularly the lack of health and location paired data.

Overall, our work demonstrates the utility of dimensionality reduction and supervised feature selection for understanding complex health data and offers a foundation for more advanced, interpretable diabetes risk modeling and geographic risk mapping in future research.

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